

ADD AND MANAGE USERS WITH LINUX COMMANDS

Authentication is the process of a user proving that they are who they say they are in the system.

When managing this, security analysts need to ensure

- not all users get access to the system,
- new users (those who are new to the organization or a group) are added to the system, and
- current users who change groups or leave the organization are deleted from the system.

Scenario

In this scenario, a new employee with the username researcher9 joins an organization. You have to add them to the system and continue to manage their access during their time with the organization.

Here's how you'll do this task: First, you'll add a new employee to the system and then to their primary group. Second, you'll make this employee the owner of a file related to a particular project. Third, you'll add the new employee to a supplementary group. Finally, you'll delete the employee from the system.

Note: The lab starts with you logged in as user analyst, with your home directory, /home/analyst, as the current working directory.

Task 1. Add a new user

A new employee has joined the Research department. In this task, you must add them to the system. The username assigned to them is `researcher9`.

1. Write a command to add a user called `researcher9` to the system.

Next, you need to add the new user to the `research_team` group.

2. Use the `usermod` command and `-g` option to add `researcher9` to the `research_team` group as their primary group.

```
analyst@59cd1940b777:~$ sudo useradd researcher9
analyst@59cd1940b777:~$ sudo usermod -g research_team researcher9
analyst@59cd1940b777:~$
```

Task 2. Assign file ownership

The new employee, `researcher9`, will take responsibility for `project_r`. In this task, you must make them the owner of the `project_r.txt` file.

The `project_r.txt` file is located in the `/home/researcher2/projects` directory, and owned by the `researcher2` user.

- Use the `chown` command to make `researcher9` the owner of `/home/researcher2/projects/project_r.txt`.

Click **Check my progress** to verify that you have completed this task correctly.

```
analyst@59cd1940b777:~$ sudo chown researcher9 /home/researcher2/projects/project_r.txt
analyst@59cd1940b777:~$
```

Task 3. Add the user to a secondary group

A couple of months later, this employee's role at the organization has changed, and they are working in both the Research and the Sales departments.

In this task, you must add `researcher9` to a secondary group (`sales_team`). Their primary group is still `research_team`.

- Use the `usermod` command with the `-a` and `-G` options to add `researcher9` to the `sales_team` group as a secondary group.

```
analyst@59cd1940b777:~$ sudo usermod -a -G sales_team researcher9
analyst@59cd1940b777:~$
```

Task 4. Delete a user

A year later, `researcher9`, decided to leave the company. In this task, you must remove them from the system.

1. Run a command to delete `researcher9` from the system:

```
sudo userdel researcher9
```



This command will output the following message:

```
oved because it is not the primary group of user researcher9.
```

This is expected.

Note: When you create a new user in Linux, a group with the same name as the user is automatically created and the user is the only member of that group. After removing users, it is good practice to clean up any such empty groups that may remain behind.

2. Run the following command to delete the `researcher9` group that is no longer required:

```
sudo groupdel researcher9
```



```
analyst@59cd1940b777:~$ sudo userdel researcher9
userdel: group researcher9 not removed because it is not the primary group of u
ser researcher9.
analyst@59cd1940b777:~$ sudo groupdel researcher9
analyst@59cd1940b777:~$
```